COURSE PLAN

(Undergraduate Programme)

(Each Teacher to submit course plan individually)

F/QSP 11/10/07

DEPARTMENT Computer Science and Information Technology

Course Code: **CT-159** Course Title: **Data Structure Algorithms and Applications**

Teacher Name: **Dr. Muhammad Kamran** Semester: **Fall** Academic Session: **2026**

Class: **2nd Year (BSCS)** No. of sections: **02** No of contact hours per week per section: **03**

ASSESSMENT METHODS

■ Assignment	■ Test/Quiz	☐ Complex Engg. Problem
■ Problem-based Learning	■ Open-ended Lab	■ Project/Group Activity
☐ Term Report/Term Paper	☐ Industrial Visit Report	☐ Case Study
☐ Poster/Other Presentation	☐ Blog/Digital Content Creation	☐ Other _____

WEEK No.	TOPICS	NO. OF PERIODS
1.	Introduction, Abstract data types, linear search, binary search	**03**
2.	Linked List - Insertion, Deletion, Search, and Sorting	**03**
3.	Doubly and Circular Linked Lists	**03**
4.	Recursion and Stack	**03**
5.	Queue, circular queue, and double-ended queue	**03**
6.	Elementary Sorting Techniques – bubble sort, insertion sort, selection sort	**03**
7.	Advanced sorting techniques – Merge sort, quick sort	**03**
8.	**Mid Term Exam**	**03**
9.	Binary Tree – Properties and its types	**03**
10.	Binary Heap, Heap sort, Priority Queue	**03**
11.	Binary Search Tree – Operations, applications, issues	**03**
12.	AVL Tree –Balance in binary search trees	**03**
13.	Graph – representation and traversal, shortest path algorithm	**03**
14.	Graph – Minimum spanning tree, graph algorithms, topological sorting	**03**
15.	Hashing – hash function, collision resolution technique, rehashing	**03**
	Total per Semester:	**45**

SESSIONAL CRITERIA

Assessment METHOD	Marks	Schedule (Week No.)
Midterm Exam	20	8th
Activity / Competition	5	10th
Assessments	10	5th , 12th
Project Report	5	15th
Total Sessional Marks	**40**	

Prepared by: Dr. Muhammad Kamran

Date 04-08-2025

Reviewed & Approved by: Dr. Muhammad Mubashir Khan

Date